

Shubham Patel

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SUMMARY

Software Developer building production backend, LLM, and multi-cloud systems. 1.5+ years full-time at Iolite Softwares; 2 years part-time during B.Tech. B.Tech CS, GPA 9.21/10. Open to SDE roles at product-led companies.

EDUCATION

Pandit Deendayal Energy University
B.Tech, Computer Science – GPA: 9.21 / 10

Gandhinagar, Gujarat
July 2021 – May 2025

EXPERIENCE

Iolite Softwares

Software Developer

January 2025 – Present

- **Agentic NL-to-SQL Chatbot** — [\[GitHub\]](#) | [\[Live\]](#): Architected a multi-tenant agentic NL-to-SQL chatbot on **LangGraph** with locally-served **Gemma**, a 3-tier follow-up engine, destructive-SQL gate, and binary-encoded RBAC whitelist across **50+ modules** – cut follow-up LLM calls by **~60%**.
- **FastSQLDatabase + RAG Retrieval**: Built a custom FastSQLDatabase wrapper with cached information_schema snapshots and multi-tier **TTL+LRU** caches; added RAG few-shot retrieval with hybrid embedding + lexical fallback – reduced tenant cold-start latency by **~70%**.
- **Hybrid Distributed Caching**: Built ETag-based hybrid HTTP caching with **IMemoryCache** and **Microsoft Garnet** on an **ASP.NET / C# / Angular 17** stack – **~70%** DB-load reduction and **~60%** faster responses.
- **Trademark Automation at Scale**: Shipped scrapers for **200+** trademark offices with custom CAPTCHA-solving models and PDF/image+text similarity pipelines (**ResNet**, **CLIP**, **RapidFuzz**) across **250+** regional journals.

TTFA Academy

Software Developer (Part-Time, during B.Tech)

January 2023 – December 2024

- **Multi-Cloud Active-Passive DR** — [\[GitHub\]](#) | [\[Blog\]](#): Designed DR topology with **Cloudflare** routing primary traffic to **AWS Mumbai**, hot-standby **AWS Singapore**, and **Azure** failover on **MongoDB Atlas**; built canary fallback (**5%** → **100%** stepped ramp) gated on multi-tier health checks for automated cutover. **Terraform** + **Jenkins** drill pipeline.
- **Adaptive Video Streaming Pipeline**: Built end-to-end HLS pipeline – **SQS**-dispatched ephemeral **EC2** workers running Dockerized **FFmpeg** transcoders, multi-bitrate (**1080p/720p/360p**) chunks served via CDN with a custom **HTML5 ABR** player to **1000+** students.
- **CRM with Lead Attribution and EMI Automation**: Built CRM with **Meta Leads** + **LinkedIn Graph** webhook ingestion, server-side **Meta Conversions API** (recovering iOS 14+ ATT attribution), and an in-house EMI engine – cut lead-response time by **~80%**.

PROJECTS

Express Docs RAG – Hybrid Search over Express.js Docs

[Live](#) | [Repo](#) | [Blog](#)

Personal Project – Python, FastAPI, ChromaDB, BM25, Cohere, Gemini, Next.js

- **Hybrid Retrieval + Reranking**: Fused **BM25Okapi** (lexical) and **ChromaDB** dense embeddings (all-MiniLM-L6-v2) via min-max-normalized weighted sum; **Cohere cross-encoder** second-stage reranking over 20–25 candidates – lifted context_precision **0.42** → **0.71**.
- **Eval CI + Provider-Agnostic Design**: Ragas metrics (faithfulness, answer relevancy, context precision/recall) gated in **GitHub Actions** block merges on regression; **Protocol** interfaces decouple LLM (**Gemini/OpenAI**), embeddings, and reranker with idle-aware model weight management.

SKILLS

Languages: Python, TypeScript, JavaScript, C#, C++, SQL

Techs: LangGraph, LangChain, RAG, ChromaDB, BM25, Cohere, Ragas, Gemini; PostgreSQL, Redis, SQL Server, multi-cloud DR, CI/CD; Tailwind CSS, REST APIs

ACHIEVEMENTS

Rank #7 / 150 – [HackThisFall 3.0](#), Feb 2023 | **Top 10** – SBSSU Hackathon, Punjab, Mar 2023

Infosys Springboard 5.0 Intern – Built [TeamSync](#), a full-stack collaborative project management platform (Node.js, Express, React, MongoDB) [\[Repo\]](#)